ECE 2510 – Intro to Microprocessors

Prelab #8

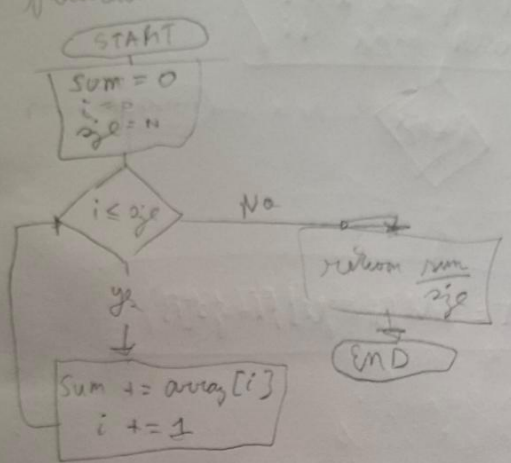
Student Name: Sebastian Manuel De Leon Santos

Sebastian De Leon

Prelab 8

Task 1

flowchart



Pseudo code

```
Process average (array, size)
    sum = 0
    i = 0
    while i < size do
        sum = sum + array[i]
        i = i + 1
    End while
    return sum / size
End process
```

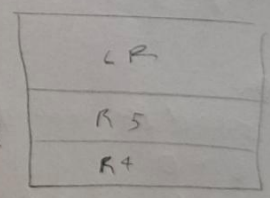
ARM Code

```
average PROC
        PUSH    {R4, R5, LR}    (3)
        MOV     R4, #0          (1)
        MOV     R5, #0          (1)
loop
        CMP     R5, R1          (1) (1)
        BGE     done            (1) (3)

        LDRSH   R2, [R0, R5 LSL #1] (2)
        ADD     R4, R2          (1)
        ADD     R5, #1          (1)
        B       loop            (3)

done
        SDIV    R0, R4, R1      (7)
        POP     {R4, R5, PC}    (4)
        ENDP
```

Cycle count

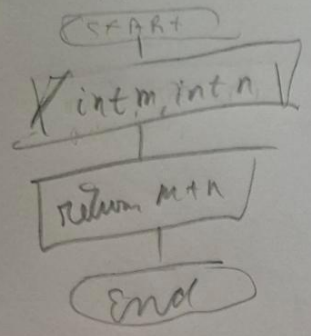
$20 + 9 \cdot N$ where N is the size of array

Stack Diagram

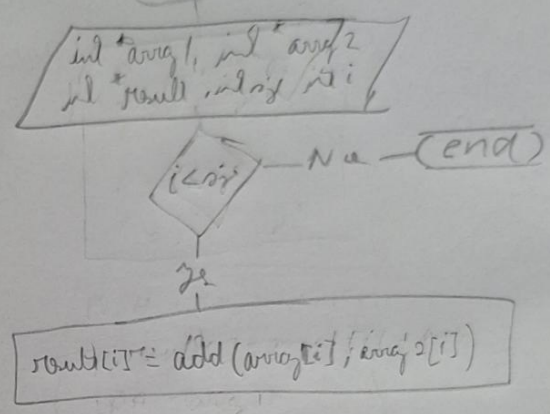


Task 2 flowcharts

add



equation



pseudocode

Main

Push size to stack
Push result array address to stack
" " array2 " "
" " array1 " "
call subroutine Equation
end program

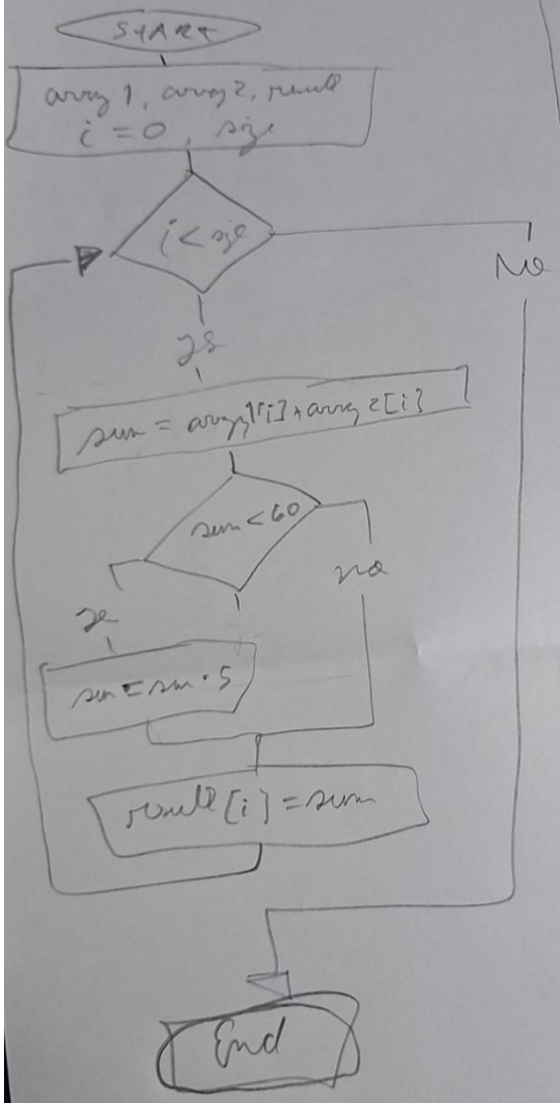
Equation

read parameters from stack
i = 0
while i < size do
reserve 4 byte on stack for return
push array2[i] and array1[i] to stack
call subroutine add
Pop returned sum from stack
remove array2[i] & array1[i] from stack
store returned sum into result[i]
i += 1;
return to main

ADD:

read m an n from stack
csum = m + n
store sum in stack
return to Equation

ask 3



Pseudo code

```
void sum (int *array1, *array2, int *result, int size)
  for i < size do:
    sum = array1[i] + array2[i]
    if sum < 60 do
      sum = sum * 5
    end if
    result[i] = sum
  end for
```